

Traffic Crash Analytics & Safety Intelligence Platform

Problem Statement:

Traffic crash data contains valuable insights that can help improve road safety, optimize emergency response, and support policy decisions. However, extracting meaningful insights from large-scale structured data requires strong SQL and analytical skills.

In this project, provided with a **pre-cleaned and structured dataset** of traffic crashes. The objective is to:

- Analyze crash data using advanced SQL techniques
- Identify patterns, trends, and risk factors
- Generate business insights from structured data

The focus is on **SQL-based problem solving, analytical thinking, and insight generation**, similar to real-world data analyst and data engineer roles.

Technologies Used:

<u>Technology</u>	<u>Purpose</u>
Python	Data Analysis
SQL	Querying
SQLite	Database
Streamlit	Dashboard
Pandas	Data Manipulation
Matplotlib/Seaborn	Visualization

Project Structure:

TRAFFIC_CRASH_PROJECT/

|

|— app.py

|— README.md

|— traffic_crash.ipynb

|— traffic_crash.sqlite

|— Traffic_CrashesData.csv

|— traffic_crash.jpg

|

|— reports/

| └─ Traffic_Crash_Project_Report.pdf

|

|— screenshots/

 |— Home Page 1.png

 |— Home Page 2.png

 |— Query Page.png

 |— Visualisation 1.png

 └─ Visualisation 2.png

Project Components:

1.Traffic_crash.ipynb

The Notebook file contains:

- Data loading and preprocessing
- SQL query execution

```
# Import Libraries
import pandas as pd
import sqlite3
```

✓ 3.0s

Python

```
# Read the Excel file using pandas
url = (r"C:\Users\harin\OneDrive\Desktop\TRAFFIC _CRASH PROJECT\Traffic_CrashesData.csv")
df = pd.read_csv(url)
```

✓ 9.9s

Python

```
pd.set_option('display.max_columns', None)
df.head()
```

✓ 0.3s

Python

	CRASH_RECORD_ID	CRASH_DATE	POSTED_SPEED_LIMIT	TRAFFIC_CONTROL_DEVICE	DEVICE_CONDITION	WEATHER_CONDITION
0	000c4307d8e9b39075cffdd0aade3603e0f96f14e41da9...	01/14/2025 12:25:00 PM	30	TRAFFIC SIGNAL	FUNCTIONING PROPERLY	SNOW
1	027b0b4c21460d3441fd83929abb9673c6fc0c7d575675...	05/23/2025 09:30:00 AM	30	STOP SIGN/FLASHER	UNKNOWN	UNKNOWN
2	04d91dff94f677358ca47056921ba5c4224320df27ed4...	04/05/2025 08:00:00 PM	30	NO CONTROLS	NO CONTROLS	CLEAR
3	0b5603954d84b7341c7cad4f570ea039e85919f3750ccb...	05/23/2025 09:15:00 AM	30	NO CONTROLS	NO CONTROLS	CLEAR
4	00bce77960c2faa2a8782a8cac1d6e5715802d6c072a08...	01/14/2025 08:00:00 AM	30	TRAFFIC SIGNAL	FUNCTIONING PROPERLY	SNOW

```
df.tail()
```

Python

	CRASH_RECORD_ID	CRASH_DATE	POSTED_SPEED_LIMIT	TRAFFIC_CONTROL_DEVICE	DEVICE_CONDITION	WEATHER_CONDI
660929	5ff5b050ae2079df6addca8431c1bd6e381432ff72ff43...	10/18/2025 11:42:00 AM	35	TRAFFIC SIGNAL	FUNCTIONING PROPERLY	CLOUDY/OVER
660930	7a7963ac15a38f3cf7549397dd053fd7612dd5194de280...	10/18/2025 03:50:00 AM	35	NO CONTROLS	NO CONTROLS	
660931	84e546c12fd8fd78ae4e2dc26fdb02d6e8f6c4c92be450...	10/18/2025 05:08:00 PM	30	NO CONTROLS	NO CONTROLS	C
660932	8237045757e7fed5adfb450bd7c8472e28a61d322f86bb...	10/19/2025 01:50:00 AM	30	UNKNOWN	UNKNOWN	UNKN
660933	ede74fa6fd507334cd0415d3d53deb535d27d23523cd38...	06/23/2023 04:30:00 PM	30	UNKNOWN	UNKNOWN	UNKN

5 rows × 39 columns

```
df.columns
```

Python

```
Index(['CRASH_RECORD_ID', 'CRASH_DATE', 'POSTED_SPEED_LIMIT',
      'TRAFFIC_CONTROL_DEVICE', 'DEVICE_CONDITION', 'WEATHER_CONDITION',
      'LIGHTING_CONDITION', 'FIRST_CRASH_TYPE', 'TRAFFICWAY_TYPE',
      'ALIGNMENT', 'ROADWAY_SURFACE_COND', 'ROAD_DEFECT', 'REPORT_TYPE',
      'CRASH_TYPE', 'DAMAGE', 'DATE_POLICE_NOTIFIED',
      'PRIM_CONTRIBUTORY_CAUSE', 'SEC_CONTRIBUTORY_CAUSE', 'STREET_NO',
      'STREET_DIRECTION', 'STREET_NAME', 'BEAT_OF_OCCURRENCE', 'NUM_UNITS',
      'MOST_SEVERE_INJURY', 'INJURIES_TOTAL', 'INJURIES_FATAL',
      'INJURIES_INCAPACITATING', 'INJURIES_NON_INCAPACITATING',
      'INJURIES_REPORTED_NOT_EVIDENT', 'INJURIES_NO_INDICATION',
      'INJURIES_UNKNOWN', 'CRASH_HOUR', 'CRASH_DAY_OF_WEEK', 'CRASH_MONTH',
      'LATITUDE', 'LONGITUDE', 'LOCATION', 'date', 'year'],
      dtype='object')
```

```
df.shape
```

Python

```
(660934, 39)
```

```
df.dtypes
```

Python

```
CRASH_RECORD_ID      object
CRASH_DATE            object
POSTED_SPEED_LIMIT    int64
TRAFFIC_CONTROL_DEVICE object
DEVICE_CONDITION       object
WEATHER_CONDITION     object
LIGHTING_CONDITION    object
FIRST_CRASH_TYPE      object
TRAFFICWAY_TYPE       object
ALIGNMENT             object
ROADWAY_SURFACE_COND  object
ROAD_DEFECT           object
REPORT_TYPE           object
CRASH_TYPE            object
DAMAGE               object
DATE_POLICE_NOTIFIED  object
PRIM_CONTRIBUTORY_CAUSE object
SEC_CONTRIBUTORY_CAUSE object
STREET_NO            int64
STREET_DIRECTION     object
STREET_NAME          object
BEAT_OF_OCCURRENCE    float64
NUM_UNITS             int64
MOST_SEVERE_INJURY    object
INJURIES_TOTAL        float64
...
LONGITUDE             float64
LOCATION              object
date                 object
year                 int64
dtype: object
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 660934 entries, 0 to 660933
Data columns (total 39 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   CRASH_RECORD_ID                       660934 non-null object
1   CRASH_DATE                             660934 non-null object
2   POSTED_SPEED_LIMIT                    660934 non-null int64
3   TRAFFIC_CONTROL_DEVICE                660934 non-null object
4   DEVICE_CONDITION                       660934 non-null object
5   WEATHER_CONDITION                     660934 non-null object
6   LIGHTING_CONDITION                    660934 non-null object
7   FIRST_CRASH_TYPE                      660934 non-null object
8   TRAFFICWAY_TYPE                       660934 non-null object
9   ALIGNMENT                             660934 non-null object
10  ROADWAY_SURFACE_COND                  660934 non-null object
11  ROAD_DEFECT                           660934 non-null object
12  REPORT_TYPE                           660934 non-null object
13  CRASH_TYPE                            660934 non-null object
14  DAMAGE                                660934 non-null object
15  DATE_POLICE_NOTIFIED                  660934 non-null object
16  PRIM_CONTRIBUTORY_CAUSE                660934 non-null object
17  SEC_CONTRIBUTORY_CAUSE                 660934 non-null object
18  STREET_NO                             660934 non-null int64
19  STREET_DIRECTION                      660934 non-null object
...
37  date                                  660934 non-null object
38  year                                  660934 non-null int64
dtypes: float64(10), int64(7), object(22)
```

```
# Create a Database in SQLite
conn_sqlite = sqlite3.connect("traffic_crash.sqlite") # Creates a local database file
cursor = conn_sqlite.cursor()
print("SQLite connection established!")
```

Python

SQLite connection established!

```
# Create a Table in SQLite
cursor.execute("""
CREATE TABLE IF NOT EXISTS crash_data (
    CRASH_RECORD_ID TEXT,
    CRASH_DATE TEXT,
    POSTED_SPEED_LIMIT BIGINT,
    TRAFFIC_CONTROL_DEVICE TEXT,
    DEVICE_CONDITION TEXT,
    WEATHER_CONDITION TEXT,
    LIGHTING_CONDITION TEXT,|
    FIRST_CRASH_TYPE TEXT,
    TRAFFICWAY_TYPE TEXT,
    ALIGNMENT TEXT,
    ROADWAY_SURFACE_COND TEXT,
    ROAD_DEFECT TEXT,
    REPORT_TYPE TEXT,
    CRASH_TYPE TEXT,
    DAMAGE TEXT,
    DATE_POLICE_NOTIFIED TEXT,
    PRIM_CONTRIBUTORY_CAUSE TEXT,
    SEC_CONTRIBUTORY_CAUSE TEXT,
    STREET_NO INTEGER,
    STREET_DIRECTION TEXT,
    STREET_NAME TEXT,
    BEAT_OF_OCCURRENCE DOUBLE,
    NUM_UNITS INTEGER,
    MOST_SEVERE_INJURY TEXT,
    INJURIES_TOTAL DOUBLE,
    INJURIES_FATAL DOUBLE,
    INJURIES_INCAPACITATING DOUBLE,
    INJURIES_NON_INCAPACITATING DOUBLE,
    INJURIES_REPORTED_NOT_EVIDENT DOUBLE,
    INJURIES_NO_INDICATION DOUBLE,
    INJURIES_UNKNOWN DOUBLE,
    -----

```

Python

```

    CRASH_HOUR BIGINT,
    CRASH_DAY_OF_WEEK BIGINT,
    CRASH_MONTH BIGINT,
    LATITUDE DOUBLE,
    LONGITUDE DOUBLE,
    LOCATION TEXT,
    date TEXT,
    year INTEGER
);
""")

conn_sqlite.commit()

print("Table 'crash_data' created successfully in SQLite!")

```

Table 'crash_data' created successfully in SQLite!

```
# Data Insertion
df.to_sql(
    "crash_data",
    conn_sqlite,
    if_exists="replace",
    index=False
)

print("Data inserted successfully!")

```

Python

Data inserted successfully!

```

query = "SELECT COUNT(*) as total_rows FROM crash_data;"
result = pd.read_sql(query, conn_sqlite)

print(result)

```

Python

```

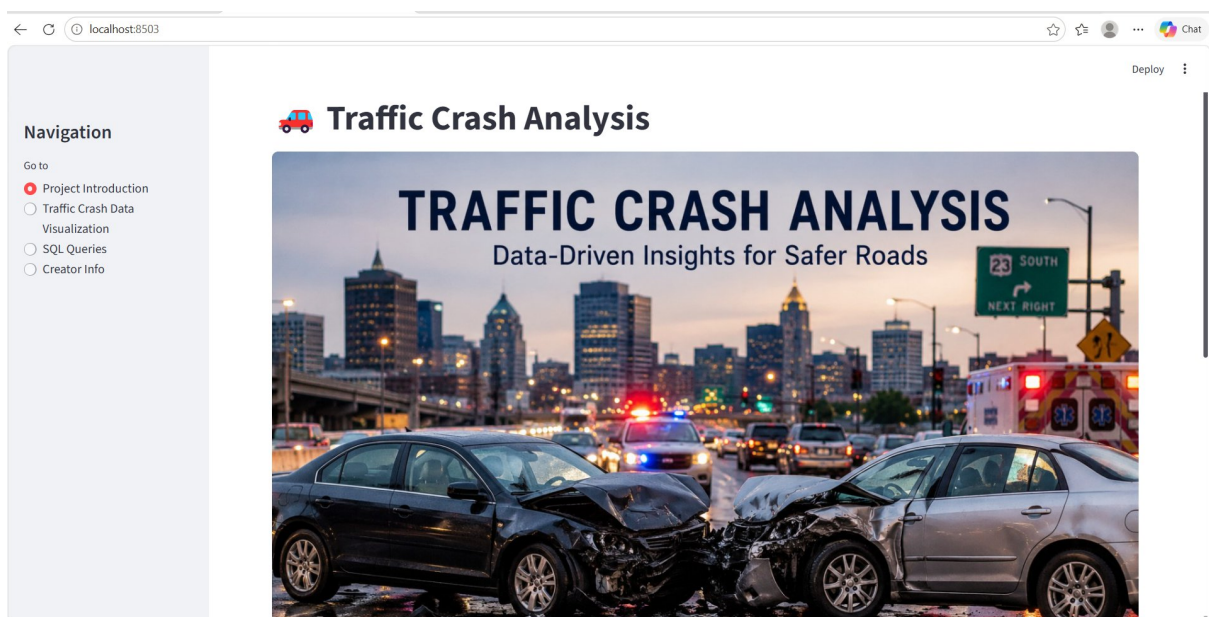
total_rows
0      660934

```

2.App.py

The Streamlit application includes:

- Interactive dashboard
- SQL query execution
- Dynamic filtering options
- Data visualization charts
- Business insight display



A Streamlit App for Analysing Traffic Crashes

This project analyzes traffic crash data using an SQLite database.

It provides insights into crash frequency, injuries, hotspot zones, weather impact, and traffic conditions through interactive visualizations and SQL analysis.

Features:

- View and analyze crash trends by year, month, day, and hour.
- Identify high-risk locations and hotspot zones.
- Explore injury patterns based on weather and lighting conditions.
- Run advanced SQL queries using CTEs and window functions to explore insights.

Database Used: `traffic_crash.sqlite`

Navigation

Go to

☐ Project Introduction

☒ Traffic Crash Data Visualization

☐ SQL Queries

☐ Creator Info

 Traffic Crash Data Visualizer

Select Street

SHERIDAN RD

Filter By:

☒ Crash Year

☐ Crash Month

☐ Weather Condition

Select Year

2020

Crash Data

	CRASH_RECORD_ID	CRASH_DATE	POSTED_SPEED_LIMIT	TRAFFIC_CONTROL_DEVICE	DEVICE_C
0	2d3a8c8cb609355fcca8312490b5740bbe4e84df00ed2fc939fabf3e950c34d3a4e69424	11/21/2020 10:36:00 AM	30	NO CONTROLS	NO CONT
1	39e6a05f1d64cfe4c11a87a73a605f5da1deb5f6e847cfd7e93c55fb6f469a7a85578633	03/09/2020 08:30:00 AM	30	NO CONTROLS	OTHER
2	3b79444419744118d1175c6a35f54c28a08130038a1bfbfc92da961b852216ef04a63d31	08/17/2020 09:20:00 AM	30	NO CONTROLS	NO CONT
3	56de8a0c6001d2e7f822978e42c4bd73e8916cbb4eb740e20191c57680a7eaf35b96865	09/15/2020 01:45:00 PM	30	NO CONTROLS	NO CONT
4	519e702161a55976ed2637b6ad958fe99d260036c31f44599a6703fabcc41192b4cbe5f	11/19/2020 04:30:00 PM	5	NO CONTROLS	NO CONT
5	bf4c3aab1bf64cf38d680a3e88fe2940479b54920d075460b20b7bc327b55e7955f1f8fc	02/15/2020 05:05:00 PM	0	NO CONTROLS	NO CONT

 Traffic Crash SQL Query Results

Choose a Query

1.Top 5 most dangerous combinations of weather and crash type

Query Result

	WEATHER_CONDITION	CRASH_TYPE	TOTAL_CRASHES
0	CLEAR	NO INJURY / DRIVE AWAY	360238
1	CLEAR	INJURY AND / OR TOW DUE TO CRASH	155302
2	UNKNOWN	NO INJURY / DRIVE AWAY	41442
3	RAIN	NO INJURY / DRIVE AWAY	31560
4	RAIN	INJURY AND / OR TOW DUE TO CRASH	19647

Business Insight: Clear weather conditions increase crash risks significantly.

"1.Top 5 most dangerous combinations of weather and crash type":

```
SELECT
    WEATHER_CONDITION,CRASH_TYPE,
    COUNT(*) AS TOTAL_CRASHES
FROM crash_data
GROUP BY WEATHER_CONDITION,CRASH_TYPE
ORDER BY TOTAL_CRASHES DESC
LIMIT 5;
```




Traffic Crash SQL Query Results

Choose a Query

1.Top 5 most dangerous combinations of weather and crash type



Query Result

	WEATHER_CONDITION	CRASH_TYPE	TOTAL_CRASHES
0	CLEAR	NO INJURY / DRIVE AWAY	360238
1	CLEAR	INJURY AND / OR TOW DUE TO CRASH	155302
2	UNKNOWN	NO INJURY / DRIVE AWAY	41442
3	RAIN	NO INJURY / DRIVE AWAY	31560
4	RAIN	INJURY AND / OR TOW DUE TO CRASH	19647

Business Insight: Clear weather conditions increase crash risks significantly.

"2. Top 10 streets with the highest number of injury crashes":
""

```
SELECT
| STREET_NAME,
| COUNT(*) AS TOTAL_INJURY_CRASHES
FROM crash_data
WHERE
| INJURIES_TOTAL > 0
GROUP BY
| STREET_NAME
ORDER BY
| TOTAL_INJURY_CRASHES DESC
LIMIT 10;
""
```



Traffic Crash SQL Query Results

Choose a Query

2. Top 10 streets with the highest number of injury crashes



Query Result

	STREET_NAME	TOTAL_INJURY_CRASHES
0	WESTERN AVE	2973
1	PULASKI RD	2840
2	ASHLAND AVE	2579
3	CICERO AVE	2560
4	HALSTED ST	2536
5	KEDZIE AVE	1965
6	STATE ST	1379
7	MICHIGAN AVE	1344
8	STONY ISLAND AVE	1340
9	DAMEN AVE	1263

Business Insight: These streets experience the highest injury-related crashes and may require stricter traffic enforcement, speed monitoring, and improved road infrastructure.

"3. Percentage of crashes that resulted in injuries for each crash type ":

"""

```
SELECT  
CRASH_TYPE,
```

```
COUNT(*) AS total_crashes,
```

```
SUM(  
  CASE
```

```
    WHEN INJURIES_TOTAL > 0 THEN 1  
    ELSE 0
```

```
  END
```

```
) AS injury_crashes,
```

```
ROUND(  
  (  
    SUM(  
      CASE
```

```
        WHEN INJURIES_TOTAL > 0 THEN 1  
        ELSE 0
```

```
      END
```

```
    ) * 100.0
```

```
  ) / COUNT(*),
```

```
  2
```

```
) AS injury_percentage
```

```
FROM crash_data
```

```
GROUP BY CRASH_TYPE
```

```
ORDER BY injury_percentage DESC;
```

"""



Traffic Crash SQL Query Results

Choose a Query

3. Percentage of crashes that resulted in injuries for each crash type

Query Result

	CRASH_TYPE	total_crashes	injury_crashes	injury_percentage
0	INJURY AND / OR TOW DUE TO CRASH	195141	103527	53.05
1	NO INJURY / DRIVE AWAY	465793	811	0.17

Business Insight: This analysis helps identify crash types with the highest injury risk, allowing emergency response teams and policymakers to prioritize safety improvements.

..... "4. Peak crash hour for each month":

```
WITH monthly_hourly_crashes AS
(
    SELECT
        CRASH_MONTH,
        CRASH_HOUR,
        COUNT(*) AS total_crashes,

        RANK() OVER(
            PARTITION BY CRASH_MONTH
            ORDER BY COUNT(*) DESC
        ) AS crash_rank

    FROM crash_data

    GROUP BY
        CRASH_MONTH,
        CRASH_HOUR
)

SELECT
    CRASH_MONTH,
    CRASH_HOUR AS peak_crash_hour,
    total_crashes

FROM monthly_hourly_crashes

WHERE crash_rank = 1

ORDER BY CRASH_MONTH;
.....
```

Choose a Query

4. Peak crash hour for each month

Query Result

	CRASH_MONTH	peak_crash_hour	total_crashes
0	1	15	4408
1	2	15	4525
2	3	15	4610
3	4	15	3884
4	5	15	4727
5	6	16	4552
6	7	16	4299
7	8	16	4504
8	9	15	4597
9	10	15	4644

Business Insight: Identifying peak crash hours helps traffic departments allocate police patrols, emergency services, and traffic management resources more effectively.

"5. Top 5 primary causes of crashes during night time (CRASH_HOUR ≥ 18) ":

```
SELECT
    PRIM_CONTRIBUTORY_CAUSE,
    COUNT(*) AS TOTAL_CRASHES
FROM crash_data
WHERE
    CRASH_HOUR >= 18
GROUP BY
    PRIM_CONTRIBUTORY_CAUSE
ORDER BY
    TOTAL_CRASHES DESC
LIMIT 5;
```

Traffic Crash SQL Query Results

Choose a Query

5. Top 5 primary causes of crashes during night time (CRASH_HOUR ≥ 18)

Query Result

	PRIM_CONTRIBUTORY_CAUSE	TOTAL_CRASHES
0	UNABLE TO DETERMINE	63606
1	FAILING TO YIELD RIGHT-OF-WAY	16213
2	FOLLOWING TOO CLOSELY	12070
3	NOT APPLICABLE	7829
4	IMPROPER OVERTAKING/PASSING	7710

Business Insight: Understanding the major causes of night-time crashes helps in creating awareness campaigns and implementing targeted safety measures during late hours.

"6. Average number of injuries in daylight vs darkness conditions":
""

```
SELECT
  LIGHTING_CONDITION,
  ROUND(AVG(INJURIES_TOTAL), 2) AS avg_injuries
FROM crash_data
WHERE LIGHTING_CONDITION IN
(
  'DAYLIGHT',
  'DARKNESS'
)
AND INJURIES_TOTAL IS NOT NULL
GROUP BY LIGHTING_CONDITION;
""
```



Traffic Crash SQL Query Results

Choose a Query

6. Average number of injuries in daylight vs darkness conditions

Query Result

	LIGHTING_CONDITION	avg_injuries
0	DARKNESS	0.21
1	DAYLIGHT	0.21

Business Insight: Comparing injuries during daylight and darkness helps evaluate the impact of visibility conditions on crash severity.

"7. Traffic control device type has the highest average injuries per crash":
""

```
SELECT
  TRAFFIC_CONTROL_DEVICE,
  ROUND(AVG(INJURIES_TOTAL),2) AS INJURIES
FROM crash_data
WHERE
  TRAFFIC_CONTROL_DEVICE IS NOT NULL
  AND INJURIES_TOTAL IS NOT NULL
GROUP BY
  TRAFFIC_CONTROL_DEVICE
ORDER BY
  INJURIES DESC
LIMIT 1;
""
```



Traffic Crash SQL Query Results

Choose a Query

7. Traffic control device type has the highest average injuries per crash

Query Result

	TRAFFIC_CONTROL_DEVICE	INJURIES
0	BICYCLE CROSSING SIGN	0.66

Business Insight: This analysis identifies traffic control systems associated with severe crashes and supports infrastructure improvement decisions.

"8. Top 5 locations (latitude/longitude) with the highest crash frequency":
""

```
SELECT
    STREET_NAME,
    LOCATION,
    LATITUDE,
    LONGITUDE,
    COUNT(*) AS crash_frequency
FROM crash_data
WHERE
    LATITUDE IS NOT NULL AND
    LONGITUDE IS NOT NULL
GROUP BY LOCATION,
    STREET_NAME,
    LATITUDE,
    LONGITUDE
ORDER BY crash_frequency DESC
LIMIT 5;
```

Choose a Query

8. Top 5 locations (latitude/longitude) with the highest crash frequency

Query Result

	STREET_NAME	LOCATION	LATITUDE	LONGITUDE	crash_frequency
0	OHARE ST	POINT (-87.905309125103 41.976201139024)	41.9762	-87.9053	1247
1	LAKE SHORE DR	POINT (-87.580147768689 41.791420282098)	41.7914	-87.5801	473
2	STONY ISLAND AVE	POINT (-87.585971992965 41.751460603167)	41.7515	-87.586	469
3	LAKE SHORE DR SB	POINT (-87.619928173678 41.900958919109)	41.901	-87.6199	376
4	STONY ISLAND AVE	POINT (-87.585275565077 41.722257273006)	41.7223	-87.5853	353

Business Insight: These hotspot locations experience frequent crashes and may require better road design, traffic signals, or monitoring systems.

"9. Top 5 streets with the highest injury rate and streets with more than 100 crashes":
""

```
WITH street_analysis AS
(
    SELECT
        STREET_NAME,
        COUNT(*) AS total_crashes,
        ROUND(AVG(INJURIES_TOTAL), 2) AS injury_rate
    FROM crash_data
    WHERE STREET_NAME IS NOT NULL
    AND INJURIES_TOTAL IS NOT NULL
    GROUP BY STREET_NAME
)

SELECT
    STREET_NAME,
    total_crashes,
    injury_rate
FROM street_analysis
WHERE total_crashes > 100
ORDER BY injury_rate DESC
LIMIT 5;
```


Choose a Query

9. Top 5 streets with the highest injury rate and streets with more than 100 crashes

Query Result

Download Search Full Screen

	STREET_NAME	total_crashes	injury_rate
0	MARQUETTE DR	271	0.45
1	FIFTH AVE	299	0.43
2	CORCORAN PL	122	0.4
3	DOUGLAS BLVD	390	0.39
4	SOUTH CHICAGO AVE	1710	0.39

Business Insight: Streets with high injury rates indicate dangerous traffic conditions and should be prioritized for road safety improvements.

"10. Identify the most common crash type for each year.":
""

WITH yearly_crash_types AS

```
(
    SELECT
        year,
        CRASH_TYPE,
        COUNT(*) AS total_crashes,

        RANK() OVER(
            PARTITION BY year
            ORDER BY COUNT(*) DESC
        ) AS crash_rank

    FROM crash_data

    GROUP BY
        year,
        CRASH_TYPE
)
```

```
SELECT
    year,
    CRASH_TYPE,
    total_crashes
```

FROM yearly_crash_types

WHERE crash_rank = 1

ORDER BY year;

""

Choose a Query

10. Identify the most common crash type for each year.

Query Result

	year	CRASH_TYPE	total_crashes
0	2020	NO INJURY / DRIVE AWAY	62706
1	2021	NO INJURY / DRIVE AWAY	74661
2	2022	NO INJURY / DRIVE AWAY	75897
3	2023	NO INJURY / DRIVE AWAY	78034
4	2024	NO INJURY / DRIVE AWAY	78560
5	2025	NO INJURY / DRIVE AWAY	77388
6	2026	NO INJURY / DRIVE AWAY	18547

Business Insight: Tracking yearly crash patterns helps authorities understand changing traffic risks and evaluate the effectiveness of safety measures over time.

"11. Day of the week with the highest average crashes per hour":

"""

```
WITH hourly_crashes AS
(
    SELECT
        CRASH_DAY_OF_WEEK,
        CRASH_HOUR,
        COUNT(*) AS total_crashes
    FROM crash_data
    WHERE CRASH_DAY_OF_WEEK IS NOT NULL
    AND CRASH_HOUR IS NOT NULL
    GROUP BY CRASH_DAY_OF_WEEK, CRASH_HOUR
),
```

daily_average AS

```
(
    SELECT
        CRASH_DAY_OF_WEEK,
        ROUND(
            AVG(total_crashes),
            2
        ) AS avg_crashes_per_hour
    FROM hourly_crashes
    GROUP BY CRASH_DAY_OF_WEEK
)
```

SELECT

CASE

CASE

```
    WHEN CRASH_DAY_OF_WEEK = 1 THEN 'Sunday'
    WHEN CRASH_DAY_OF_WEEK = 2 THEN 'Monday'
    WHEN CRASH_DAY_OF_WEEK = 3 THEN 'Tuesday'
    WHEN CRASH_DAY_OF_WEEK = 4 THEN 'Wednesday'
    WHEN CRASH_DAY_OF_WEEK = 5 THEN 'Thursday'
    WHEN CRASH_DAY_OF_WEEK = 6 THEN 'Friday'
    WHEN CRASH_DAY_OF_WEEK = 7 THEN 'Saturday'
```

END AS day_name,

avg_crashes_per_hour

FROM daily_average

ORDER BY avg_crashes_per_hour DESC

LIMIT 1;

""

Choose a Query

11. Day of the week with the highest average crashes per hour

Query Result

	day_name	avg_crashes_per_hour
0	Friday	4455.17

Business Insight: Identifying high-risk weekdays helps optimize staffing, traffic monitoring, and public safety planning.

"12. Group hours into buckets (Morning, Afternoon, Evening, Night)for high-risk time slots and bucket with highest injury crashes":

""

WITH time_bucket_analysis AS

(

SELECT

CASE

WHEN CRASH_HOUR BETWEEN 6 AND 11
 THEN 'Morning'

WHEN CRASH_HOUR BETWEEN 12 AND 16
 THEN 'Afternoon'

WHEN CRASH_HOUR BETWEEN 17 AND 20
 THEN 'Evening'

ELSE 'Night'

END AS time_bucket,

SUM(INJURIES_TOTAL) AS total_injuries

FROM crash_data

GROUP BY time_bucket

)

SELECT

time_bucket,
 total_injuries

FROM time_bucket_analysis

ORDER BY total_injuries DESC

LIMIT 1;

""



Traffic Crash SQL Query Results

Choose a Query

12. Group hours into buckets (Morning, Afternoon, Evening, Night)for high-risk time slots and bucket with highest injury crashes

Query Result

	time_bucket	total_injuries
0	Afternoon	44239

Business Insight: High-risk time slots indicate when severe crashes occur most frequently, helping emergency teams prepare resources during critical hours.

"13. Top 3 contributing causes for each crash type":
""

SELECT

CRASH_TYPE,
PRIM_CONTRIBUTORY_CAUSE,
COUNT(*) AS total_crashes

FROM crash_data

WHERE PRIM_CONTRIBUTORY_CAUSE IS NOT NULL

GROUP BY

CRASH_TYPE,
PRIM_CONTRIBUTORY_CAUSE

ORDER BY total_crashes DESC

LIMIT 3;
""

Choose a Query

13. Top 3 contributing causes for each crash type

Query Result

	CRASH_TYPE	PRIM_CONTRIBUTORY_CAUSE	total_crashes
0	NO INJURY / DRIVE AWAY	UNABLE TO DETERMINE	206391
1	INJURY AND / OR TOW DUE TO CRASH	UNABLE TO DETERMINE	66377
2	NO INJURY / DRIVE AWAY	FOLLOWING TOO CLOSELY	45964

Business Insight: Identifying major contributing causes helps traffic authorities implement targeted prevention strategies and driver awareness programs.

"14. Year-over-year growth rate of crashes":
""

WITH yearly_crashes AS

```
(
  SELECT
    year,
    COUNT(*) AS total_crashes
  FROM crash_data
  WHERE year IS NOT NULL
  GROUP BY year
),
```

growth_analysis AS

```
(
  SELECT
    year,
    total_crashes,

    LAG(total_crashes) OVER
    (
      ORDER BY year
    ) AS previous_year_crashes

  FROM yearly_crashes
)
SELECT
  year,
  total_crashes,
  previous_year_crashes,

  ROUND(
    (
      (total_crashes - previous_year_crashes)
      * 100.0
    ) / previous_year_crashes,
    2
  ) AS growth_rate_percentage
```

FROM growth_analysis

WHERE previous_year_crashes IS NOT NULL

ORDER BY year;

""

Choose a Query

14. Year-over-year growth rate of crashes

Query Result

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	year	total_crashes	previous_year_crashes	growth_rate_percentage
0	2021	107937	91509	17.95
1	2022	107462	107937	-0.44
2	2023	109714	107462	2.1
3	2024	110853	109714	1.04
4	2025	107965	110853	-2.61
5	2026	25494	107965	-76.39

Business Insight: Crash growth trends help evaluate whether traffic safety initiatives are reducing or increasing accident rates over time.

"15. Group nearby locations (round latitude & longitude to 2 decimal places) and Find top 10 zones with highest crashes":

WITH hotspot_zones AS

```
(
    SELECT
        ROUND(LATITUDE, 2) AS latitude_zone,
        ROUND(LONGITUDE, 2) AS longitude_zone,
        COUNT(*) AS total_crashes

    FROM crash_data

    WHERE LATITUDE IS NOT NULL
    AND LONGITUDE IS NOT NULL

    GROUP BY
        ROUND(LATITUDE, 2),
        ROUND(LONGITUDE, 2)
),
```

zone_rank AS

```
(
    SELECT
        latitude_zone,
        longitude_zone,
        total_crashes,

        RANK() OVER
        (
            ORDER BY total_crashes DESC
        ) AS crash_rank

    FROM hotspot_zones
)

SELECT
    latitude_zone,
    longitude_zone,
    total_crashes,
    crash_rank

FROM zone_rank

WHERE crash_rank <= 10

ORDER BY crash_rank;
"""
```

Choose a Query

15. Group nearby locations (round latitude & longitude to 2 decimal places) and Find top 10 zones with highest crashes

Query Result

	latitude_zone	longitude_zone	total_crashes	crash_rank
0	41.89	-87.63	7633	1
1	41.88	-87.63	5552	2
2	41.89	-87.62	5550	3
3	41.9	-87.62	5049	4
4	41.88	-87.62	4272	5
5	41.88	-87.64	3951	6
6	41.9	-87.63	3935	7
7	41.99	-87.66	3392	8
8	41.89	-87.64	3158	9
9	41.87	-87.64	3111	10

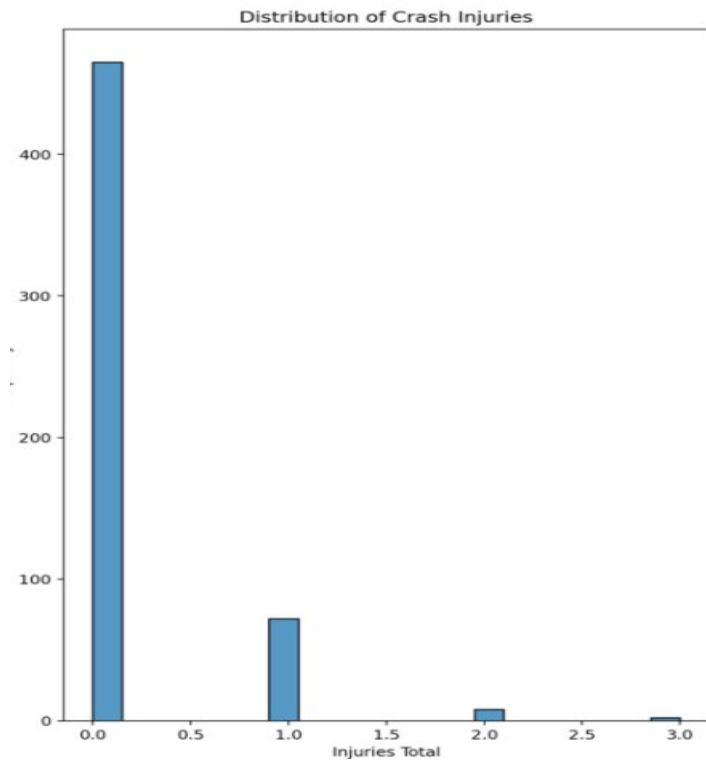
Business Insight: High-crash zones highlight areas requiring urgent traffic safety interventions, improved infrastructure, and accident prevention measures.

Key Features:

- Traffic crash trend analysis
- Injury severity analysis
- Weather impact analysis
- High-risk street identification
- Crash hotspot zone detection
- Time-based crash analysis
- Interactive Streamlit dashboard
- Advanced SQL analytics

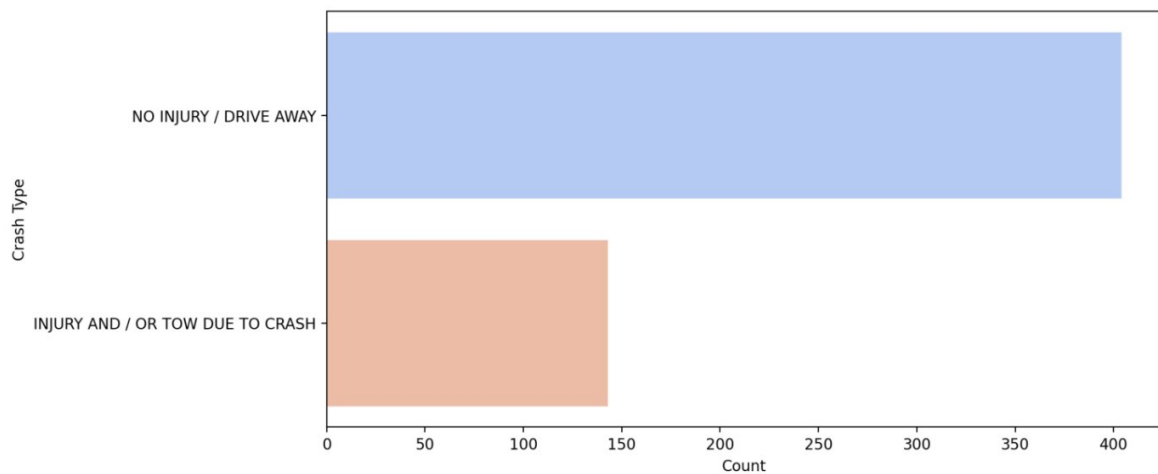
Visualizations Used

- Line Charts



- Count Plots

Crash Type Distribution



Business Insights Generated:

- Identified dangerous weather and crash type combinations
- Detected high-risk streets with maximum injuries
- Analyzed peak crash hours and dangerous time slots
- Compared daylight and darkness injury severity
- Identified hotspot crash zones using geographic coordinates
- Found yearly crash trends and growth rates

Conclusion:

The project successfully transformed raw traffic crash data into meaningful insights using SQL, Python, SQLite, and Streamlit. Interactive dashboards and analytical queries helped identify important crash patterns, dangerous locations, and contributing factors that can support better road safety and decision-making.

Author:

A. Harini

Skills:

- Python
- SQL
- Streamlit
- Data Analysis
- Pandas
- Data Visualization